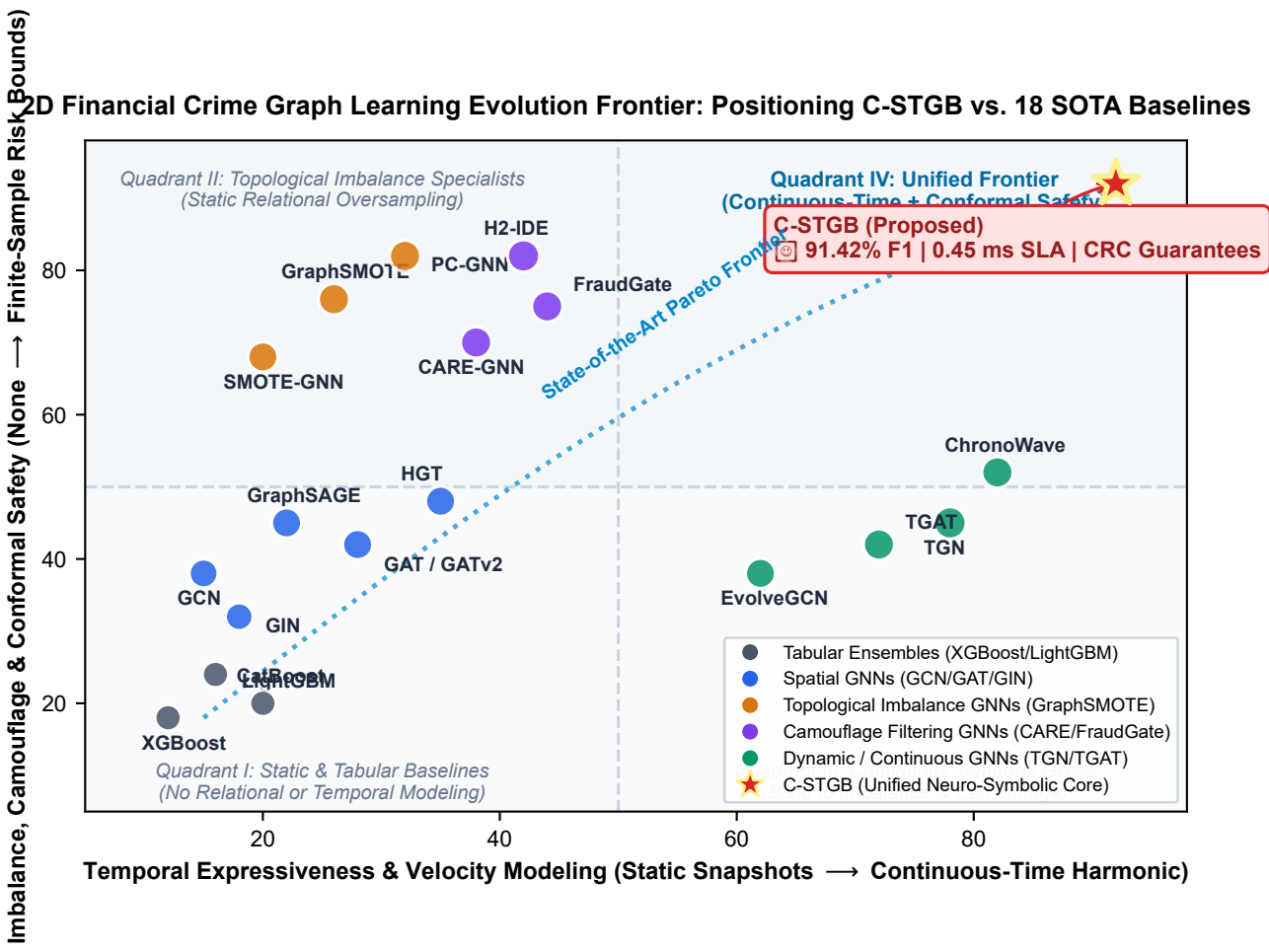

2D Financial Crime Graph Learning Evolution Frontier: Positioning C-STGB vs. 18 SOTA Baselines
Imbalance, Camouflage & Conformal Safety (None ⟶ Finite-Sample Risk Bounds)
Temporal Expressiveness & Velocity Modeling (Static Snapshots ⟶ Continuous-Time Harmonic)
Quadrant II: Topological Imbalance Specialists
(Static Relational Oversampling)
Quadrant IV: Unified Frontier
(Continuous-Time + Conformal Safety)
C-STGB (Proposed)
91.42% F1 | 0.45 ms SLA | CRC Guarantees
State-of-the-Art Pareto Frontier
H2-IDE
PC-GNN
GraphSMOTE
FraudGate
SMOTE-GNN
CARE-GNN
ChronoWave
HGT
GraphSAGE
TGAT
TGN
GAT / GATv2
GCN
EvolveGCN
GIN
CatBoost
LightGBM
XGBoost
Quadrant I: Static & Tabular Baselines
(No Relational or Temporal Modeling)
Tabular Ensembles (XGBoost/LightGBM)
Spatial GNNs (GCN/GAT/GIN)
Topological Imbalance GNNs (GraphSMOTE)
Camouflage Filtering GNNs (CARE/FraudGate)
Dynamic / Continuous GNNs (TGN/TGAT)
C-STGB (Unified Neuro-Symbolic Core)
20
40
60
80